

Bachelor Applied Data Science & Artificial Intelligence

Course Name	Calculus
Course Code	PR12
Performance Indicator	1
Academic Year	2025–2026
Block period	4
Test / Resit	Mock Test
Test Date	N/A
Start Time – End Time	N/A
Separate answer sheet	No
Examiners	Baljan, J. & Alberts, F.
Maximum Number of Points (T)	40
Randomly Expected Points (E)	0
Passing Threshold Points	22

Number of Points (P)	
Student Name	
Student Identification Number	

Calculation PI-grade: $(P-E)/(T-E)*10$.

The examination board can make a generic correction to this after test analysis.

Instructions

- Allowed aids: writing utensils, pen with black or blue ink
- Check the entire set for completeness and call the exam supervisor if parts are missing
- Complete the entries in the spaces indicated on this form and/or on the answer sheet
- Hand everything in to the exam supervisor afterwards.

Question 1 (2 points)

Consider the function

$$f(x) = \begin{cases} \frac{x^2 - 4x}{x - 4}, & x \neq 4, \\ a, & x = 4. \end{cases}$$

Determine the value of a for which f is continuous at $x = 4$.

Question 2 (3 points)

Consider the function

$$g(x) = \begin{cases} 3x^2 + b, & x < 2, \\ 6, & x = 2, \\ bx + 18, & x > 2. \end{cases}$$

Determine all values of b for which g is continuous at $x = 2$.

Question 3 (4 points)

Consider the function

$$h(x) = \sin(2x)e^{\sin(2x)}$$

Find $h'(x)$

Question 4 (4 points)

Given the function

$$f(x) = \frac{1}{3}x^3 + x^2 - 8x - 5$$

Find the critical points of f , classify them using the first or second derivative test, and determine the interval where f is concave up and concave down.

Question 5 (6 points)

Compute

$$\int_0^2 x^2 e^x dx$$

Question 6 (4 points)

Evaluate

$$\int \frac{3x^2}{x^3 + 1} dx$$

Question 7 (5 points)

Construct the degree 3 Maclaurin polynomial of

$$f(x) = \sin 2x$$

Question 8 (6 points)

Solve the initial value problem

$$y' + 4y = e^{-3x}, \quad y(0) = 2$$

Question 9 (6 points)

Find the maximum and minimum values of

$$f(x, y) = -2x + 2y$$

subject to the constraint

$$x^2 + y^2 = 3.$$